Hojun Jeong

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RESEARCH INTERESTS

Knowledge Editing; Sequential and Continual Learning; Catastrophic Forgetting; Post-Training Knowledge Dynamics; Representation Dynamics and Mechanistic Interpretability

EDUCATION

Gachon University - M.S. in AI Software, Artificial Intelligence Mar. 2025 - Present
Advisor: Prof. Sangwoo Kang | GPA: 4.3 / 4.5
• **Research Focus:** Knowledge Editing, Continual Learning, Catastrophic Forgetting, and Representation Dynamics

Gachon University - B.S. in AI Software, Artificial Intelligence Mar. 2023 - Feb. 2025
GPA: 3.6 / 4.5

Dongguk University - Coursework in Computer Engineering Mar. 2018 - Feb. 2023
Transferred to Gachon University

PUBLICATIONS

Manuscript Under Review

***Hojun Jeong**, *Gyunyeon Kim, and Sangwoo Kang. "**Target-Token Prior as a Predictor of Forgetting in Fine-Tuning-Based Knowledge Editing**." ACL Rolling Review, May 2026 cycle. Under review.

Conference Proceedings

***Hojun Jeong**, *Gyunyeon Kim, and Sangwoo Kang. "**KLOD: Locality-Preserving Knowledge Editing via Non-Target Distribution Preservation**." Findings of the Association for Computational Linguistics: EMNLP 2026. To appear.

* *Equal contribution.*

Hojun Jeong and Sangwoo Kang. "**A Verify-Refine Hybrid for Dialogue Summarization with Topic Segmentation and Keypoint Coverage**." In Proceedings of the 37th Annual Conference on Human and Language Technology (HCLT 2025), pp. 359-364, October 2025. *[Paper in Korean]*

Hojun Jeong and Sangwoo Kang. "**Structured Zero-Shot Prompts through Rule-Based Intent Analysis for Dialogue Summarization**." In Proceedings of the Korea Computer Congress (KCC 2025), July 2025. *[Paper in Korean]*

RESEARCH EXPERIENCE

Graduate Research Student, NLP / AI Lab, Gachon University Mar. 2025 - Present
Advisor: Prof. Sangwoo Kang

KLOD: Locality-Preserving Knowledge Editing via Non-Target Distribution Preservation

- Identified an objective-level mismatch in standard cross-entropy editing: target probability can increase while non-target token distributions drift, causing accumulated locality degradation under sequential updates.
- Proposed KLOD, combining bounded target logit-odds optimization with KL preservation of the target-excluded non-target distribution and the full next-token distribution at prefix positions.
- Led method development, implementation, baseline selection, experimental design, and ablation studies across two backbones and two benchmarks, including 3,000-edit sequential protocols and scaling experiments up to 150,000 edits.
- On LLaMA3-8B-Instruct/CounterFact, improved Locality from **1.87 to 44.75** while maintaining **99.70 Reliability**; on ZsRE, achieved 86.47 Locality with 99.89 Reliability. Distribution-KL analyses identified target-excluded non-target preservation as the main source of locality gains.

Target-Token Prior as a Predictor of Forgetting in Fine-Tuning-Based Knowledge Editing

- Extended an initial diagnostic observation into a controlled synthetic biography study with 50,000 pre-training and 50,000 fine-tuning individuals across five prior-based bins; target-token NLL separated forgetting trajectories more consistently than generic output confidence.
- Proposed PLFT-KE, a two-stage prior-aware warm start that optimizes a latent probability lift on a frozen model, materializes it as a rank-1 parameter initialization, and consolidates the edit with standard fine-tuning.
- Led post-diagnosis method development, implementation, baseline selection, experimental design, and analysis on CounterFact and ZsRE with Qwen2.5-7B and LLaMA-3.2-1B/3B.

- On LLaMA-3.2-3B/CounterFact, improved overall Score from **58.43 to 72.41**, Generalization from 41.75 to 57.83, and Specificity from 33.55 to 59.40; gains reached 31.4 specificity points in the lowest-prior bin. Stress-tested 2,000 sequential edits, identifying accumulated-update limits that motivated KLOD.

Ongoing Research: Early-Stage Forgetting and Layer-Wise Transformation

- Investigating why severe forgetting is concentrated in the earliest stages of fine-tuning, including under distribution-preserving objectives.
- Analyzing layer-wise transformations from rotational and scaling perspectives to connect representation subspaces, predictive distributions, and computational pathways used to express knowledge.

Research Assistant, NLP / AI Lab, Gachon University Sep. 2023 - Feb. 2025

Advisor: Prof. Sangwoo Kang

- Conducted research on dialogue summarization and knowledge editing for large language models; implemented fine-tuning, prompting, and evaluation pipelines using PyTorch and Hugging Face Transformers.
- Contributed to manuscript preparation, research presentations, and regular laboratory seminars.

AWARDS AND HONORS

Outstanding Presentation Paper Award, HCLT 2025, October 2025. Awarded for "A Verify-Refine Hybrid for Dialogue Summarization with Topic Segmentation and Keypoint Coverage."

PRESENTATIONS

"A Verify-Refine Hybrid for Dialogue Summarization with Topic Segmentation and Keypoint Coverage." Oral Presentation, HCLT 2025, October 2025.

"Structured Zero-Shot Prompts through Rule-Based Intent Analysis for Dialogue Summarization." Poster Presentation, KCC 2025, July 2025.

PROFESSIONAL SKILLS

Programming: Python; Java and C (working familiarity)

ML & Data Analytics: PyTorch; TensorFlow; Keras; Pandas; NumPy; Matplotlib; Hugging Face Transformers; LangChain; EasyEdit

Platforms & Tools: Git; Linux; LaTeX

Research Methods: Supervised and Localized Fine-Tuning; Knowledge Editing; Sequential and Continual Learning; KL-Regularized Training; Token-Level Probability and Output-Distribution Analysis; Ablation Study Design; Language-Model Evaluation

Languages: Korean (native); English (academic reading and writing)